

HELIOS 100 MW Solar PV and 33/220kV Substation: Engineering and Design Management Framework

1. Engineering Department Overview

The engineering department within an Engineering, Procurement, and Construction (EPC) turnkey framework operates under fundamentally different commercial and technical pressures compared to a traditional Design-Bid-Build (DBB) model. In a traditional DBB environment, the design is fully matured by the client's consultant before tendering, isolating the designer from construction execution risks. Under a FIDIC Yellow Book EPC contract, the contractor assumes absolute liability for the design, integration, and ultimate performance of the facility¹. The engineering department thus functions as the central risk management hub. Every technical decision requires a simultaneous evaluation of material costs, supply chain lead times, and constructability. The team must optimize the plant architecture to maximize the Internal Rate of Return (IRR) while ensuring strict compliance with local statutory codes, such as the Uzbek KMK building standards and UzbekEnergo grid requirements.

The transition from conceptual engineering to field execution is governed by a strict, formalized document maturation process, culminating in the Issued for Construction (IFC) milestone. This process ensures that no procurement or construction activities commence without verified and approved documentation. The progression of engineering deliverables follows a distinct nomenclature:

- **IFR (Issued for Review):** Documents at this stage represent a mature draft submitted for internal interdisciplinary coordination or preliminary client feedback. The structural logic is complete, but final vendor data (such as specific transformer dimensions or inverter weights) may still be pending.
- **IFA (Issued for Approval):** The document is finalized by the EPC engineering team, incorporating all interdisciplinary checks, and is formally transmitted to the Employer or the Lender's Independent Engineer (IE). Under standard FIDIC Yellow Book Sub-Clause 5.2 provisions, the Employer is typically granted a review period not exceeding 21 days².
- **IFC (Issued for Construction):** A document achieves IFC status only after the Employer or IE returns the IFA submittal with an "Approved" or "Approved as Noted" code. The IFC stamp transforms the drawing into a legally binding instrument. Site execution and sub-contractor works are permitted solely from IFC-stamped documents.
- **As-Built:** Following the completion of construction, the engineering team updates the native IFC files to reflect the precise physical reality of the installed plant. This incorporates all red-line markups from the field, resolved Requests for Information (RFIs), and Field Design Changes (FDCs).

The Design Manager orchestrates this vast technical operation. Operating at the intersection of engineering, procurement, and project management, the Design Manager's daily routine is

characterized by constraint resolution and critical-path protection. A typical day involves mediating interdisciplinary clashes—such as resolving a spatial conflict between a medium-voltage underground cable trench designed by the electrical team and a perimeter security fence foundation designed by the civil team. The Design Manager constantly monitors the Master Document Register (MDR) against the Primavera P6 schedule, prioritizing the review of vendor submittals that unblock long-lead manufacturing slots, and facilitating high-level technical negotiations with the off-taker, UzbekEnergo, regarding grid interconnection methodologies.

2. Drawing Register & Document Control

The sheer volume of technical documentation generated during a 100 MW utility-scale solar project necessitates an uncompromising document control architecture. The foundational element of this architecture is the Master Document Register (MDR) or Drawing Register. The MDR is a dynamic database that dictates the planned and actual lifecycle of every engineering deliverable.

The MDR establishes a sequence of accountability. Maintaining the MDR involves weekly updates integrated with the project schedule, allowing the project controls team to calculate the earned value of engineering progress. Parallel to the MDR is the Submittal Register, which tracks Vendor Document Requirements (VDR). While the MDR tracks documents authored by the EPC, the Submittal Register tracks documents provided by equipment manufacturers (e.g., tracker aeroelastic reports, transformer factory acceptance test protocols). Tracking vendor documents is critical because engineering cannot issue IFC foundation drawings until the vendor supplies certified equipment weights and baseplate dimensions.

Formal communication of these documents is executed via Document Transmittals. A transmittal is a legally binding cover letter that accompanies the transfer of documents, establishing a contractual timestamp. It details the transmittal number, date, sender, recipient, purpose of issue (e.g., For Approval, For Construction), and a schedule of the enclosed documents with their specific revision codes. This timestamp is vital for defending against or substantiating claims related to review period delays under the FIDIC contract².

Document numbering conventions and revision structures in solar EPCs are highly codified to prevent fatal errors on site. Utilizing outdated revisions can result in thousands of dollars in rework. A standard document code follows a rigid syntax, such as HEL-CV-LAY-0010-RevB:

- HEL: Project Identifier (HELIOS).
- CV: Discipline Code (Civil).
- LAY: Document Type (Layout).
- 0010: Sequential Identifier.
- RevB: Revision Code.

Revisions utilizing alphabetic characters (A, B, C) denote pre-construction phases (IFR, IFA). Once a document achieves IFC status, the revision transitions to numeric characters (0, 1, 2). Thus, Rev 0 always denotes the first IFC issuance, and Rev 1 denotes a subsequent post-IFC design change.

A critical commercial milestone in this lifecycle is the Design Freeze. Achieved after the preliminary design phase, the design freeze locks the macroeconomic parameters of the plant:

total module count, inverter capacities, structural pitch, and the geographic footprint of the 220kV substation. Freezing these parameters allows the Supply Chain Management (SCM) team to execute multimillion-dollar purchase orders for long-lead items. Any alteration proposed after the design freeze—such as altering the string configuration which would impact the DC combiner box internal busbars—requires a formal Management of Change (MOC) process to evaluate the cascading impacts on cost, schedule, and system performance.

3. Civil & Structural Engineering

Civil and structural engineering for a utility-scale solar plant in Uzbekistan requires navigating a complex matrix of semi-arid environmental conditions, significant seasonal temperature variations, and high seismic activity. The civil deliverables package for a 100 MW project is extensive, encompassing overall site grading plans, cut-and-fill optimization models, internal access road profiles, storm water drainage networks, module mounting structure (MMS) foundation coordinates, inverter station foundation details, perimeter fencing, and the complete 220kV substation civil infrastructure (control building architecture, transformer oil containment pits, gantry structural details, and buried earthing grid trenches).

The foundation of all civil design is the geotechnical investigation. In Uzbekistan, this investigation must align with national building codes, specifically KMK 2.02.01-98 (Foundations of Buildings and Structures) or its updated equivalent ShNK 2.02.01-19⁴. The geotechnical campaign involves boreholes, dynamic cone penetration testing, and electrical resistivity testing. Critical parameters extracted for pile design include soil bearing capacity, stratigraphy, aggressiveness of groundwater (sulfate/chloride content dictating cement type), and the maximum frost depth. Given the extreme winters in the region, foundation embedment must exceed the frost line to prevent frost heave from physically pushing the tracker piles out of the ground⁶. Furthermore, soil resistivity data is directly fed to the electrical team to design the IEEE 80 compliant earthing grid.

Simultaneously, the topographic survey establishes the macro-geometry of the site. Utilizing drone photogrammetry paired with RTK GPS ground control points, the survey generates a Digital Elevation Model (DEM) with tight contour intervals (0.25m). The DEM is imported into grading optimization software. The civil lead (TERRA) works with the tracker engineer (SOLARIS) to balance the earthworks. Trackers have specific slope tolerances (typically up to 15% N-S and 10% E-W). If the natural terrain exceeds these limits, the terrain must be graded, generating costly earthworks.

Tracker pile foundations (typically galvanized steel W-sections or C-channels) are designed to withstand a combination of dead loads, live loads (snow), and lateral aerodynamic loads (wind). The design parameters are dictated by KMK 2.01.07-96 (Loads and Impacts), which defines the regional wind velocity pressures and snow loads⁷. Pre-construction pile testing is mandatory. The EPC drives test piles and subjects them to axial compression, axial tension (pull-out), and lateral load tests to verify the geotechnical assumptions. Failure modes include axial pull-out due to aerodynamic uplift, lateral yield causing the torque tube to bind and overload the slew drive motor, and buckling during ramming into unforeseen rocky strata.

Seismic considerations are paramount. Uzbekistan features high seismicity, with regions rated at 8 or 9 on the MSK-64 scale. Structural design must strictly adhere to KMK 2.01.03-19

(Construction in Seismic Areas)⁹. While flexible tracker structures naturally dissipate seismic energy, rigid, heavy assets like the 100 MVA power transformers require specialized foundation engineering, including enhanced reinforcement detailing in the concrete plinths and vibration-isolating anchorages¹⁰.

Once construction begins, the civil team manages the As-Built survey. Surveyors use RTK rovers to capture the exact X, Y, and Z coordinates of every driven pile. These coordinates are cross-referenced against the theoretical design coordinates. Common civil RFIs revolve around pile refusal. If a pile hits a subsurface boulder at 1.0 meters instead of the designed 1.5 meters, the pile cannot be driven further without buckling. The engineering resolution typically involves issuing a Field Design Change (FDC) to extract the pile, pre-drill the hole, insert the pile, and backfill with a weak concrete mix or structural epoxy.

4. MMS / Tracker System Engineering

The Module Mounting Structure (MMS) is the mechanical skeleton of the power plant. The selection between Fixed-Tilt structures and Single-Axis Trackers (SAT) fundamentally alters the plant's yield profile and structural risks. Fixed-tilt systems are structurally robust, statically determinate, and require minimal maintenance. However, SATs dynamically pivot the PV modules to follow the sun's trajectory from east to west, increasing the total energy yield by 15% to 20%.

This increased yield introduces severe dynamic vulnerabilities. Single-axis trackers are highly susceptible to wind-induced aeroelastic instabilities. The most critical failure mode is torsional galloping—a self-exciting phenomenon where aerodynamic forces lock into phase with the natural torsional frequency of the tracker array, causing the torque tube to twist violently and exponentially until catastrophic structural failure occurs¹².

To mitigate this, tracker vendor submittals undergo intense scrutiny. The vendor must provide static structural calculations, dynamic wind tunnel test reports (proving the efficacy of their stow-strategy and mechanical dampers), motor duty cycle analyses, and certification against IEC 62817 (Design Qualification of Solar Trackers)¹⁵. The IEC 62817 standard validates the tracker's pointing accuracy, torsional stiffness, drive torque, and survivability under extreme environmental conditions, serving as the benchmark for bankability¹⁷.

The engineering deliverables for the MMS package center on the Tracker Layout Drawing. This document maps the precise location of every tracker row, the placement of the slew drives, the routing of the auxiliary power supply to the motors, and the locations of the meteorological stations required to trigger emergency stow procedures during high wind events.

The erection sequence is highly regimented: piles are driven, bearing housings are bolted to the piles, the continuous torque tube is threaded through the bearings, purlins are attached, and finally, the PV modules are clamped. Torque specifications for every bolted connection are critical to structural integrity. Vendors specify exact torque values (e.g., 65 Nm for module clamps), which must be applied using calibrated torque wrenches and marked with torque-seal paint for quality control auditing. The Bill of Materials (BOM) generated by the tracker engineer (SOLARIS) feeds directly into procurement, detailing the exact quantities of piles, torque tubes, slew drives, dampers, bearing assemblies, and millions of specific fasteners.

Installation issues frequently generate Non-Conformance Reports (NCRs). The most common

issue is pile misalignment. If adjacent piles are driven out of tolerance (horizontally or vertically), forcing the torque tube into the bearings induces pre-stress and high friction. This forces the tracking motor to operate above its rated current, leading to premature failure and tracking inaccuracies. Resolving this requires RFIs to approve the use of custom adapter brackets or the complete redriving of the out-of-tolerance piles.

5. DC Electrical Engineering

The DC electrical system is responsible for the collection and aggregation of power from the hundreds of thousands of photovoltaic modules. The DC design package comprises the DC Single Line Diagram (SLD), string sizing calculations, string layout drawings, underground cable trenching plans, inverter station electrical layouts, DC cable schedules, and voltage drop calculations.

The fundamental calculation in DC engineering is string sizing, which determines how many modules are connected in series. The critical constraint is the maximum system voltage limit of the inverter and the modules (typically 1500V DC). Because the open-circuit voltage (V_{oc}) of a PV module has a negative temperature coefficient (voltage rises as temperature falls), the engineer must calculate the string voltage at the historical absolute minimum ambient

temperature of the site (e.g., $-25^{\circ}C$ in Uzbekistan). If the string is too long, the extreme cold will push the voltage above 1500V, violating the equipment ratings and IEC 60364-7-712 installation standards¹⁹. Conversely, the string must be long enough so that the voltage at the maximum operating temperature does not drop below the inverter's Maximum Power Point Tracking (MPPT) voltage window.

The string configuration dictates the sizing of the Combiner Boxes (e.g., 20 or 24 input fuses) and the cross-sectional area of the DC cabling. The DC Cable Schedule is a meticulously detailed database used by the installation teams. It defines the point-to-point logic of the plant.

Table_title: DC Cable Schedule Parameters

Parameter	Description
Cable ID	Unique tag (e.g., CB01-STR15)
Source	Originating point (e.g., Tracker Row 45, String A)
Destination	Terminating point (e.g., Combiner Box 12, Input 15)
Specification	Conductor material and size (e.g., 6mm ² Cu, PV1-F 1500V)

Length	Estimated routing length (used for voltage drop validation)
Routing Method	Aerial along torque tube / Direct buried in sand bedding
Polarity	Positive (Red) / Negative (Black)

DC earthing in modern 1500V plants utilizing transformerless inverters generally operates as an ungrounded (floating) IT system on the DC side to minimize leakage currents. However, safety mandates that all exposed conductive parts (MMS steel, module frames, combiner box enclosures) be bonded to a continuous buried equipotential earth conductor, ultimately tied back to the inverter station earthing ring.

Common DC RFIs arise during cable trenching. Rocky soil may prevent the trenching machine from reaching the specified 800mm depth. The DC engineer (AMPERE) must evaluate if a shallower trench is permissible by adding mechanical protection (concrete tiles or thicker conduit) and recalculating the cable ampacity derating due to the altered thermal resistivity of the shallower burial. As-Built DC drawings are generated post-installation by surveying the exact as-laid route of the main DC feeder cables and recording the final string-to-combiner box mapping, which often deviates slightly from design due to field routing efficiencies.

6. AC / Substation Engineering

The AC electrical infrastructure performs the critical task of stepping up the power from the medium voltage (MV) output of the inverters to the high voltage (HV) transmission level required by the off-taker. For the HELIOS project, this involves a 33kV internal collection grid and a 220kV export switchyard.

The full AC design package encompasses the 33kV MV Ring Main SLD, the 33/220kV Substation SLD, comprehensive Substation Equipment Layouts (plan and section views), Substation Earthing Layouts, Lightning Protection Plans, MV/HV Cable Schedules, and the Relay/Control Room Architectural Layouts.

A central engineering responsibility is the preparation of equipment specifications. The main step-up transformer is the most expensive and longest-lead asset on the project. For a 100 MW solar plant, the transformer is typically specified at 100 MVA or 125 MVA to allow headroom for reactive power support¹¹. The specification will dictate the voltage ratio (220/33kV), cooling classifications (ONAN/ONAF, dictating natural vs. forced air cooling), impedance voltages (typically between 10% and 15% to limit fault currents), and the vector group (e.g., YNd11, indicating a wye-connected high voltage side and a delta-connected low voltage side with a 30-degree phase shift)²¹.

The MV switchgear specifications dictate the use of Vacuum Circuit Breakers (VCBs) rated for the system voltage (36kV maximum equipment rating for a 33kV system) and the calculated

short-circuit withstand capacity (e.g., 31.5 kA for 3 seconds)²³. Current Transformers (CTs) and Potential Transformers (PTs) are specified with exact ratios (e.g., 600/5A for CTs) and accuracy classes to feed the protection relays and revenue meters²⁴.

Grid connection design requires intense coordination with UzbekEnergo. The EPC must submit Load Flow studies to prove the plant can inject 100 MW without causing overvoltage at the Point of Common Coupling (PCC). Short Circuit studies must prove that the fault current contribution from the PV inverters does not exceed the breaking capacity of the utility's existing equipment. The physical safety of the substation is ensured through the Substation Earthing Design, governed strictly by IEEE 80²⁵. The objective is to design a buried grid of bare copper conductors that limits the Ground Potential Rise (GPR) during a fault, ensuring that personnel touching equipment (Touch Voltage) or walking across the yard (Step Voltage) are not subjected to lethal currents²⁶. The permissible touch voltage is highly dependent on the surface layer. Engineering specifies a 100mm to 150mm layer of high-resistivity crushed rock (e.g.,

$\rho_s = 3000 \Omega \cdot m$) across the switchyard²⁷. This creates high contact resistance at the feet.

The IEEE 80 equation calculates the derating factor (C_s) and the subsequent tolerable touch voltage (E_{touch}) based on body weight (typically 50kg or 70kg) and fault clearing time (t_s)²⁸.

$$C_s = 1 - \frac{0.09 \left(1 - \frac{\rho}{\rho_s}\right)}{2h_s + 0.09}$$

$$E_{touch50} = \frac{(1000 + 1.5C_s\rho_s) \times 0.116}{\sqrt{t_s}}$$

If the calculated touch voltage exceeds the tolerable limit, the engineer (VOLTA) must iterate the design by tightening the grid spacing or adding deeper ground rods until safety margins are achieved²⁷.

7. Protection & SCADA Engineering

Protection engineering safeguards the high-value assets of the plant and ensures grid stability, while the SCADA system provides the operational intelligence necessary to run the facility.

The Protection Philosophy document establishes the logic for isolating faults. Modern substations utilize microprocessor-based relays that analyze waveforms from the instrument transformers. Key protection schemes in a 220kV solar substation include:

- **87L (Line Current Differential):** Utilizes Kirchhoff's current law. Relays at both ends of the 220kV transmission line communicate via fiber optics. If the current entering the line does not equal the current leaving, an internal fault is detected, and the breakers trip instantaneously³⁰.
- **87T (Transformer Differential):** Protects the main step-up transformer. It defines a protection zone between the 220kV and 33kV CTs, tripping the transformer if an internal

winding short occurs³².

- **21 (Distance Protection):** Measures impedance ($Z = V/I$) to detect faults on the transmission line. It acts as the primary backup to the 87L scheme.
- **64N (Restricted Earth Fault):** Highly sensitive protection against earth faults near the neutral point of the transformer windings.

The setting calculation process requires a dynamic model of the grid to determine the exact thresholds (setpoints) at which these relays will actuate, ensuring they operate quickly but selectively—meaning only the breaker closest to the fault trips, preserving the rest of the plant. SCADA (Supervisory Control and Data Acquisition) in a modern PV plant utilizes dual communication architectures. The substation Intelligent Electronic Devices (IEDs) communicate via IEC 61850, a protocol optimized for high-speed, peer-to-peer relay communications (GOOSE messaging). Downstream, the field inverters and meteorological stations typically communicate over a fiber-optic ring network using Modbus TCP. SCADA monitors critical data points: active power (MW), reactive power (MVar), DC string currents, tracker angles, switchgear status, and meteorological data.

To comply with IEC 61724-1:2021 Class A monitoring requirements, the meteorological stations must utilize high-accuracy pyranometers equipped with active heating and ventilation to prevent frost and dew accumulation, ensuring the energy yield models are compared against highly accurate irradiance data³⁴.

Grid code compliance with UzbekEnergo requires sophisticated control logic executed by the Power Plant Controller (PPC). Under modern grid codes, Inverter-Based Resources (IBRs) must provide Primary Frequency Response (PFR)³⁶. If the grid frequency rises (indicating an oversupply of power on the national grid), the SCADA system must autonomously execute a droop control curve, curtailing the active power output of the solar plant proportionally to the frequency deviation to help stabilize the grid³⁸.

Before deployment, the protection and SCADA panels undergo a Factory Acceptance Test (FAT)⁴⁰. The engineer (CIPHER) travels to the panel builder's facility to witness the injection of simulated secondary currents into the relays, proving the trip logic. Upon delivery to the site, a Site Acceptance Test (SAT) is performed, involving full loop checks from the primary HV equipment, through the copper control wiring, and up to the SCADA Human-Machine Interface (HMI) screens to ensure end-to-end functionality.

8. Design Coordinator Role

The Design Coordinator (NEXGEN) operates as the logistical engine of the engineering department. While discipline leads (TERRA, AMPERE) focus on technical calculations, the coordinator focuses on process adherence, information flow, and schedule protection.

A typical day involves auditing the Submittal Register. If a tracker vendor submits their aeroelastic wind tunnel report, the coordinator receives it, logs it, assigns it a tracking code, and routes it to the structural lead (TERRA) for foundation load reviews and to the design manager (ARCHON) for commercial compliance. The coordinator polices the internal Service Level Agreements (SLAs), ensuring the engineering team returns comments to the vendor within the agreed timeframe (e.g., 10 days) so that manufacturing slots are not lost.

When construction begins, the coordinator manages the RFI (Request for Information) process. Site queries—such as encountering buried debris during trenching—must be formalized via an RFI. The coordinator logs the RFI, numbers it sequentially, and assigns it to the relevant engineer with a strict SLA (typically 3 days). A delayed RFI response can halt a subcontractor's work front, leading to costly standby claims against the EPC.

During the final phases of the project, the design coordinator manages the As-Built compilation process. This involves aggregating all red-line markups from the site electrical and civil supervisors, verifying that all approved FDCs and RFIs have been drafted into the final CAD files, and assembling the hundreds of Operation and Maintenance (O&M) manuals into the final handover package for the client.

9. Real Document Templates

To ensure completeness and accuracy in project execution and AI agent simulation, the following standardized templates define the structure of critical engineering control documents.

Table 1: Drawing Register (Master Document Register) - Sample Extract

Docu ment Num ber	Docu ment Title	Disci pline	Rev	Statu s	Plan ned IFR	Actu al IFR	Plan ned IFA	Actu al IFA	IFC Date
HEL- CV-L AY-0 001	Over all Site Gradi ng & Drain age Plan	Civil	B	IFA	12-F eb-26	14-F eb-26	01-M ar-26	02-M ar-26	Pendi ng
HEL- CV-L AY-0 002	Acce ss Road Profil es & Detail s	Civil	A	IFR	15-F eb-26	15-F eb-26	05-M ar-26	Pendi ng	Pendi ng
HEL- ST-C AL-0	Track er Pile	Struc tural	0	IFC	10-Ja n-26	10-Ja n-26	05-F eb-26	04-F eb-26	28-F eb-26

010	Geotechnical Calculations								
HEL-EE-SLD-0020	DC Single Line Diagram (Overall)	DC Elec	B	IFA	20-Feb-26	21-Feb-26	15-Mar-26	16-Mar-26	Pending
HEL-EE-SCH-0021	DC Cable Schedule	DC Elec	A	IFR	22-Feb-26	22-Feb-26	18-Mar-26	Pending	Pending
HEL-EE-LAY-0030	Inverter Station 1 Equipment Layout	DC Elec	C	IFA	01-Mar-26	02-Mar-26	20-Mar-26	Pending	Pending
HEL-HV-SLD-0100	33kV MV Ring Main SLD	AC Elec	0	IFC	15-Jan-26	15-Jan-26	10-Feb-26	11-Feb-26	05-Mar-26
HEL-HV-SLD-0	220kV Switchyard	AC Elec	B	IFA	25-Feb-26	25-Feb-26	20-Mar-26	20-Mar-26	Pending

101	Single Line Diagram								
HEL-HV-LAY-0110	Substation Equipment Layout Plan	AC Elec	B	IFA	28-Feb-26	01-Mar-26	22-Mar-26	Pending	Pending
HEL-HV-CAL-0120	Earth ing Grid Calc (IEEE 80)	AC Elec	0	IFC	05-Jan-26	06-Jan-26	01-Feb-26	30-Jan-26	20-Feb-26
HEL-HV-SCH-0130	MV AC Cable Schedule	AC Elec	A	IFR	10-Mar-26	11-Mar-26	30-Mar-26	Pending	Pending
HEL-PR-PHL-0200	Protection & Control Philosophy	Protection	0	IFC	15-Dec-25	16-Dec-25	15-Jan-26	14-Jan-26	10-Feb-26
HEL-PR-CAL-0	Relay Setting Calcul	Protection	B	IFA	01-Mar-26	03-Mar-26	25-Mar-26	Pending	Pending

201	lations (220kV)								
HEL-SC-ARC-0300	SCADA Network Architecture Diagram	SCADA	0	IFC	20-Jan-26	20-Jan-26	15-Feb-26	14-Feb-26	01-Mar-26
HEL-SC-LST-0301	SCADA I/O Point List	SCADA	A	IFR	15-Mar-26	Pending	05-Apr-26	Pending	Pending
HEL-ME-LAY-0400	Meteorological Station Layout (IEC 61724)	SCADA	0	IFC	01-Feb-26	01-Feb-26	25-Feb-26	24-Feb-26	15-Mar-26
HEL-ST-LAY-0500	Tracker Row and Slew Drive Layout	Structural	B	IFA	10-Mar-26	12-Mar-26	30-Mar-26	Pending	Pending

HEL-CV-LAY-0501	Substation Buried Trenching Plan	Civil	B	IFA	12-Mar-26	Pending	02-Apr-26	Pending	Pending
HEL-HV-SPC-0600	100MVA 220/33kV Power Transformer Spec	AC Elec	0	IFC	05-Dec-25	05-Dec-25	10-Jan-26	09-Jan-26	01-Feb-26
HEL-EE-SPC-0601	DC Combiner Box Technical Specification	DC Elec	0	IFC	15-Dec-25	16-Dec-25	20-Jan-26	20-Jan-26	10-Feb-26

Table 2: Request for Information (RFI) Form

Field	Data Entry
Project Name:	HELIOS 100 MW Solar PV + 33/220kV GSS
RFI Number:	RFI-CIV-0042
Date Raised:	15-May-2026

Raised By / Company:	John Doe / Earthworks Subcontractor Ltd.
Discipline:	Civil / Structural
Reference Document:	HEL-CV-LAY-0001-Rev0 (Grading Plan)
Subject:	Unforeseen bedrock encountered at Inverter Station 4
Query/Description:	Excavation for Inverter Pad 4 encountered solid limestone bedrock at 0.6m depth. IFC design requires 1.2m depth for frost protection per KMK 2.02.01-98. Excavator unable to reach depth. Request permission to alter foundation design to avoid blasting.
Response Required By:	18-May-2026 (To avoid standby charges)
Engineering Response:	Bedrock negates the risk of frost heave. Proceed to clear all loose topsoil down to the bedrock layer. Dowel 4x 16mm rebars 300mm into the bedrock using Hilti RE-500 structural epoxy. Cast the concrete pad directly onto the cleaned rock surface. An updated FDC sketch is attached. Rev 1 of the layout drawing will follow.
Responded By / Role:	TERRA (Civil & Structural Lead Engineer)
Date Answered:	16-May-2026

Table 3: Document Transmittal Form

Field	Data Entry
Transmittal No:	TRN-EPC-CLI-0089
Date of Issue:	21-Mar-2026

From:	ARCHON (EPC Design Manager)
To:	Client / Lender's Independent Engineer
Purpose of Issue:	Issued for Approval (IFA)
Contractual Remarks:	Submitted pursuant to FIDIC Yellow Book Sub-Clause 5.2. The 21-day review period commences upon receipt. Target return date for comments is 11-Apr-2026. If no response is received, documents will be deemed approved and issued IFC.
Enclosures:	1. HEL-HV-SLD-0101 (Rev B) - 220kV Switchyard SLD 2. HEL-HV-LAY-0110 (Rev B) - Substation Equipment Layout 3. HEL-HV-CAL-0120 (Rev 0) - Earthing Grid Calculation

Table 4: Submittal Register (Vendor Documents)

Submittal No	Vendor	Equipment	Document Title	Rev	Status	Date Sent	Target Return	Actual Return	Status Code
SUB-TRF-001	ABB	100M VA TX	General Arrangement Drawing	1	Closed	15-Jan-26	25-Jan-26	25-Jan-26	Code B
SUB-TRF-	ABB	100M VA	Rating Plate	2	Closed	28-Jan-26	07-Feb-26	02-Feb-26	Code A

002		TX	& Nameplate						
SUB-SAT-015	NEX Tracker	SAT System	Wind Tunnel Aeroelastic Report	1	Open	10-Mar-26	20-Mar-26	Pending	N/A
SUB-INV-042	SMA	Inverter	Modbus Register Map	1	Closed	05-Feb-26	15-Feb-26	14-Feb-26	Code C

Table 5: Design Review Checklist (Prior to IFC Issuance)

Verification Item	Discipline	Verified By	Date	Status
Client/IE comments from IFA phase fully incorporated	Coordinator	NEXGEN	10-Mar-26	PASS
Geotechnical bearing limits verified against structural foundation loads	Structural	TERRA	11-Mar-26	PASS
Clash detection	Interdisciplinary	ARCHON	12-Mar-26	PASS

cleared (e.g., Civil trenches vs. UG cables)				
Vendor certified data (weights, dimensions) match layout	Electrical	VOLTA	12-Mar-26	PASS
Constructability review signed off by Site Manager	Construction	Site Mgr	14-Mar-26	PASS
Safety in Design (SiD) risk matrix updated and mitigated	Safety	HSE Lead	14-Mar-26	PASS

Table 6: As-Built Drawing Index (Handover Package Structure)

Volume	Section	Contents Description
Volume 1	Project Management	Signed Contract, Permits, Grid Connection Agreements, Statutory Approvals.
Volume 2	Civil & Structural	As-Built Grading Plans, Topographic Verifications, Final Pile Coordinates (X,Y,Z), Road Profiles, Fence Layouts.
Volume 3	Electrical (DC & AC)	As-Built SLDs, Final Cable Schedules, True Trench Routings, Earthing

		Resistance Test Reports, Inverter Commissioning Certificates.
Volume 4	Protection & SCADA	Final Relay Setting Files, SCADA Architecture As-Built, Point-to-Point Testing Sign-offs, FAT/SAT Reports.
Volume 5	O&M Manuals	Vendor Operational Manuals, Maintenance Schedules, Warranty Certificates, Spare Parts Lists, IEC 62817 Tracker Certificates.

10. Engineering Interfaces

Engineering execution relies on deep logistical integration with surrounding project departments. The output of the design team directly fuels procurement, planning, and quality control.

With Supply Chain Management (SCM): Engineering translates drawings into the Material Take-Off (MTO). The MTO is a highly structured database feeding the SCM Enterprise Resource Planning (ERP) system. For example, a DC electrical MTO will not merely list "cable"; it specifies "Cable, Solar, 1-core, 6mm², Cu, XLPO/XLPO, 1500VDC, UV-Resistant, Black, 250,000 meters." Engineering must lock the MTO early enough to accommodate long shipping lead times to Uzbekistan.

With Planning: The planning department relies on engineering to establish the drawing release schedule. This is built via reverse-engineering the construction critical path. If the construction schedule dictates that tracker pile driving begins in Month 4, the IFC pile coordinates must be on site by Month 3. This means IFA submission must occur in Month 2 to allow for the 21-day review period. The Design Manager (ARCHON) owns the delivery of this schedule.

With Construction: Construction interacts continuously with engineering through the RFI and FDC processes. As field conditions deviate from desktop assumptions, the site raises RFIs. SLAs dictate that engineering must respond within 3 to 5 days. If a trenching crew is halted due to a clash, every hour of delay costs the EPC money. Engineering must provide technically sound, rapidly executable solutions.

With Quality: The Quality Assurance (QA) team relies on engineering to define the mathematical acceptance criteria within the Inspection and Test Plan (ITP). Engineering dictates the tolerances (e.g., pile horizontal deviation ± 50 mm, tracker bolt torque $65 \text{ Nm} \pm 5\%$).

Quality inspectors use these engineering-defined parameters to sign off on hold points during construction.

With Vendors: Vendor drawings are reviewed by engineering to ensure the purchased equipment fits the plant's macro-design. The review cycle results in status codes:

- **Code A (Approved):** The document is flawless. The vendor may proceed with manufacturing.
- **Code B (Approved with Comments):** The vendor may manufacture, provided they incorporate the minor red-line markups (e.g., moving a terminal block 10mm to the left).
- **Code C (Revise and Resubmit):** Fundamental errors exist. The vendor must correct and resubmit. Manufacturing is heavily blocked.
- **Code D (Rejected):** The submittal is completely non-compliant with the technical specification.

With Client / Independent Engineer (IE): Engineering serves as the technical diplomats of the EPC. Under FIDIC Yellow Book, all Contractor's Documents must be reviewed by the Employer or their IE². The contract strictly defines a Review Period, typically 21 days³. If the IE rejects a drawing (Code C), the 21-day clock resets upon resubmission. If the IE fails to respond within 21 days, the EPC may issue a formal notice of deemed approval and proceed to IFC to protect the project schedule.

11. Common Engineering Problems

The mark of a robust engineering team is its ability to adapt to dynamic, real-world constraints without compromising safety, commercial viability, or contract compliance.

Vendor Submits Non-Compliant Tracker Design If a tracker vendor submits a structural package lacking a comprehensive aeroelastic flutter analysis, the structural lead (TERRA) must immediately issue a Code C (Revise and Resubmit) status. Because torsional galloping can destroy a plant¹³, the Design Manager instructs SCM to freeze all milestone payments to the vendor until certified wind tunnel test data and IEC 62817 compliance certificates are produced¹⁵.

Geotechnical Report Shows Soil Conditions Different from Tender During the tendering phase, soil may have been assumed to be deep, loose sand. However, post-contract geotechnical investigations reveal a dense layer of calcarenite rock at 1.0 meter. Standard driven piles will buckle upon impact. The engineering team must quickly redesign the foundation matrix, pivoting to pre-drilling or concrete micro-piles. Commercially, the Design Manager assists the contract administration team in filing a claim under FIDIC Sub-Clause 4.12 (Unforeseeable Physical Conditions) to recover the increased costs and time associated with the heavier civil works¹.

Utility Changes Protection Relay Settings After Design is Frozen If UzbekEnergo updates its grid code midway through the project, altering the Fast Frequency Response (FFR) droop curve requirements³⁶, the protection engineer (CIPHER) must intercept the panels before they leave the factory. They evaluate if the new logic can be achieved via a software firmware update to the Power Plant Controller (PPC), or if the physical CT burdens and hardware must be replaced. If hardware changes are required, the EPC submits a formal Variation Order (VO) to the client to cover the redesign and remanufacturing costs.

Drawing Issued IFC Has an Error Found on Site A site excavation team discovers a clash: the IFC drawing shows a 33kV direct-buried cable running straight through a newly installed concrete drainage culvert. The construction manager raises an urgent RFI. The Design Coordinator (NEXGEN) fast-tracks this to the civil and electrical leads. Within hours, the engineers issue a Technical Query (TQ) response instructing the site to lower the cable trench profile by 600mm to pass safely underneath the culvert, maintaining thermal spacing. The electrical lead subsequently updates the drawing and issues IFC Revision 1 to formalize the fix.

Client Engineer Requests Major Layout Change After IFC Occasionally, after IFC drawings are issued and trenching has commenced, the Client's IE may request a major alteration, such as relocating the 220kV substation 50 meters north to avoid a newly discovered cultural heritage site. The Design Manager immediately issues a Stop Work Order for the affected zone. This triggers the Variation Order (VO) process. Engineering calculates the total impact: scrapped copper cable, aborted civil earthworks, and the man-hours required to redraw the layouts, earthing grids, and cable schedules. Redesign work strictly does not commence until the Client formally signs the VO, agreeing to the cost and schedule extensions.

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